

Ryan M. Burns

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PERSONAL SUMMARY

Experienced Data Scientist with a proven ability to design, train, and evaluate custom state-of-the-art genAI models and agentic apps leveraging distributed HPC resources to scale deep learning models for production. Strong background driving projects through the full data science lifecycle and delivering insightful narratives to clients.

PROGRAMMING & TECHNICAL SKILLS

Project Highlights

More projects & code available at ryanmburns93.github.io

Generative AI Healthcare Pathway Modeling

- Trained a GPT-2 language model on health condition event sequences to learn underlying disease progression pathways. Customized tokenizer and generative models for tasks including sequence classification for high-cost claimant classification, provider rank ordering, and at-risk population analysis.

Health Insurance Claims Denial Neural Network

- Designed and deployed custom transformer and XGBoost model architectures targeting insurance claim denials and reason codes. Achieved high performance while ensuring model interpretability and bias mitigation, and productionalized models for real-time inference on client claims using MLFlow and AWS.

Lease Renewal Modeling

- Modeled the likelihood an apartment resident will renew their lease by renewal offer price and connected leasing and operational cost models to optimize pricing and deliver recommendations to property managers.

Languages

- **Python** (Expert): Agentic AI w/ LangChain, NLP w/ PyTorch & TensorFlow, HuggingFace model customization, data mining/processing, production app builds w/ FastAPI, Pydantic, TinkerUI
- **SQL** (Expert): Query and data manipulation in RDBMS, NoSQL, and graph databases
- **Scala** (Expert): Dataset and feature design, model pipeline training and QA testing
- **Cloud DevOps** (Intermediate): AWS s3 data storage & SageMaker modeling, Azure serverless app build

PROFESSIONAL EXPERIENCE

Certilytics, Leading healthcare predictive analytics partner improving patient outcomes **Mar 2023 – May 2026**

Data Scientist

- Applied cutting-edge AI, ML, and NLP techniques to build client-tailored models predicting member health outcomes, system utilization, and cost savings for more than 100 conditions and over 30M patients weekly.
- Designed, scaled, and automated ML-Ops processes for model suite retraining, QA testing, and monitoring.
- Delivered model results and actionable insights directly to clients, serving as a subject-matter expert and trusted partner to advance product adoption and identify new projects and additional business opportunities.

AIR Communities, Fourth-largest U.S. apartment REIT with 96 communities in 12 states **Nov 2021 – Mar 2023**

Innovation & Automation Associate

- Designed predictive models for business stakeholders as the project lead through the full data science life cycle. Automated data collection, cleansing, and storage, applied AI/ML tools and measured model performance, and deployed and maintained models in production using DevOps best practices.
- Reduced database migration QA project staffing by 15 employees via scripting automated validation tests, constructed Power Apps for visitor record and maintenance ticket continuity during system downtime.

Gartner, Inc., World's leading research advisory company with annual revenue of \$4B **Aug 2017 – Oct 2021**

Client Success Partner, Associate Client Partner, Account Management Specialist

- Diagnosed top challenges of ~150 C-suite finance, procurement, and real estate executives representing a client territory of \$4.5M contract value. Tailored client experiences delivering relevant benchmark research.

EDUCATION

Northwestern University, Evanston, IL

Master of Science, June 2022

- M.S. in Data Science, Artificial Intelligence Specialization (GPA: 3.67)

Duke University, Durham, NC

Bachelor of Arts, September 2017

- B.A. in Public Policy Studies, Minor in Political Science (GPA: 3.35)